Homework 3 Summary

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SPCS224

1. Purpose

The purpose of this homework is to add new features to the Farkle Game. I added Full House, single-player mode, multiplayer mode, and a final round.

2. New Features

I added a new dice combination called Full House. It uses five dice with three of the same number and two of another number. It gives 1500 points.

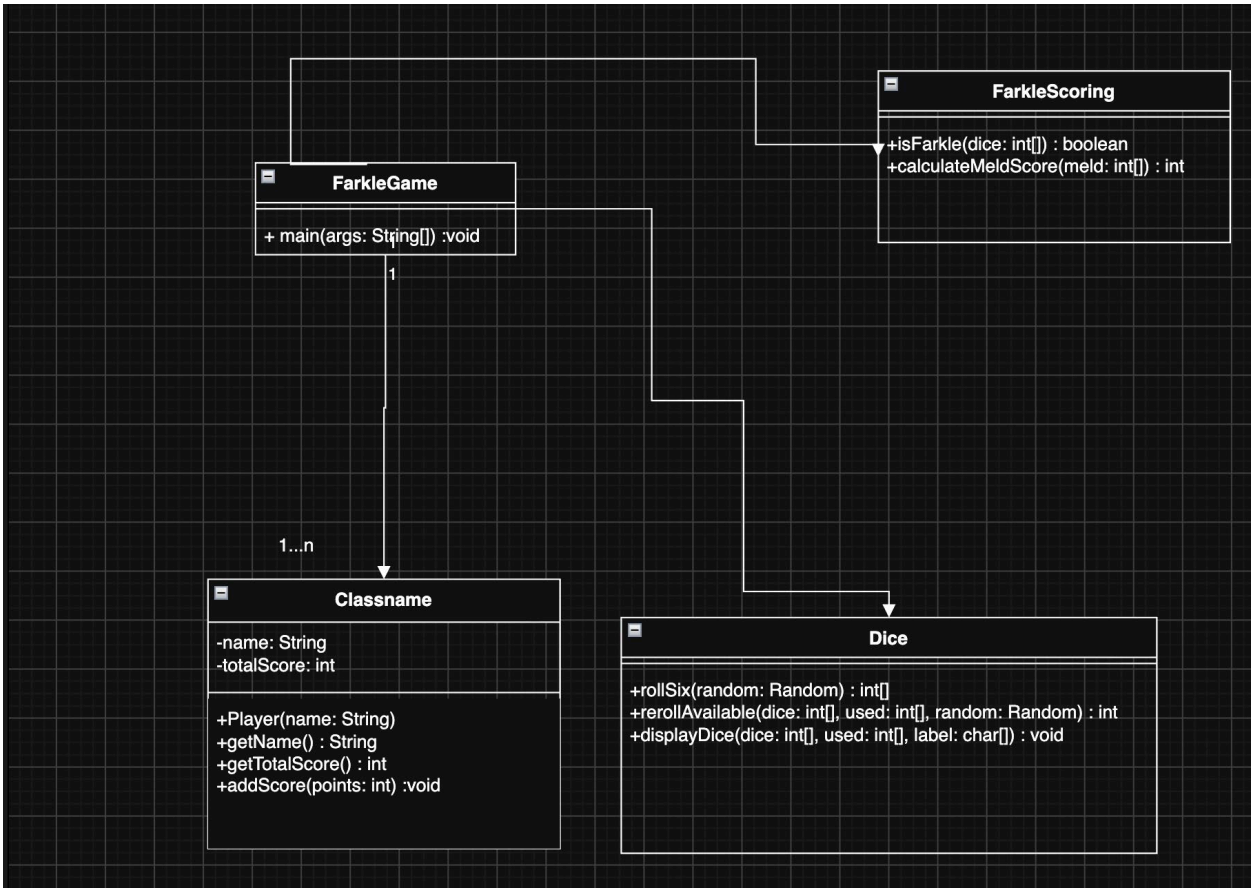
I also added single-player and multiplayer modes. Players can choose the target score and enter their names. In multiplayer mode, each player takes turns playing the game.

When a player reaches the target score, every other player gets one final turn. After that, the game shows all scores and announces the winner.

3. UML Diagrams and Screenshots

I included a UML class diagram and a UML sequence diagram to show the program structure and how the game works. I also added screenshots of Full House, single-player mode,

multiplayer mode, and the final results.



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|   Farkle Game by [Yin Ru]    |
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How many points would you like to play in this game? [10000]
1
How many players?
1
What is player 1's name? [Baby]

================================
Turn 1
Player: Baby
Total Score: 0
Target Score: 1
================================
```

```
Current Dice:
A: 5 [Available]
B: 5 [Available]
C: 1 [Available]
D: 1 [Available]
E: 1 [Available]
F: 5 [Available]
Choose dice to meld (A-F), "K" to bank, "R" to reroll the dice, or "Q" to quit:
a,b,c,d,e
Your choice is: a,b,c,d,e
New meld score: 1500
Turn score: 1500
Current Dice:
A: 5 [Meld]
B: 5 [Meld]
C: 1 [Meld]
D: 1 [Meld]
E: 1 [Meld]
F: 5 [Available]
Choose dice to meld (A-F), "K" to bank, "R" to reroll the dice, or "Q" to quit:
k
Your choice is: k
Score banked: 1500
Total score: 1500
```

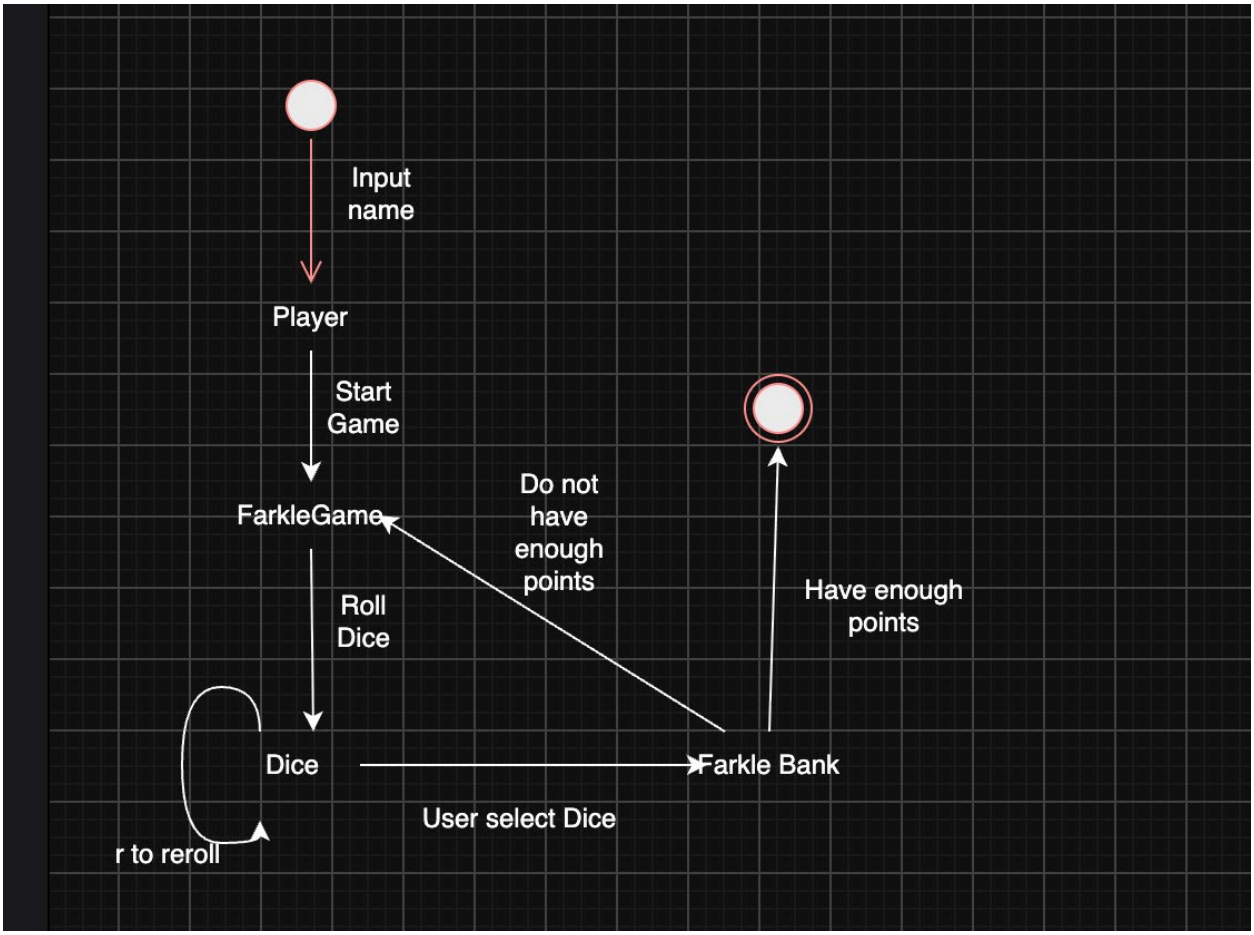
```
Current Total Scores:
Baby: 1500

==============================
Winner: Baby
Congratulations!!!
Final total score:
Baby: 1500
==============================
```

```
Current Total Scores:
Baby: 2100
Scary: 200
Final Round! Each player gets one more turn.
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Turn 6
Player: Scary
Total Score: 200
Target Score: 1000
================================
Current Dice:
A: 1 [Available]
B: 1 [Available]
C: 1 [Available]
D: 1 [Available]
E: 2 [Available]
F: 2 [Available]
Choose dice to meld (A-F), "K" to bank, "R" to reroll the dice, or "Q" to quit:
```

```
Current Total Scores:
Baby: 2100
Scary: 1800

==============================
Winner: Baby
Congratulations!!!
Final total score:
Baby: 2100
Scary: 1800
==============================
```



4. Unit Testing

I used JUnit to test the new features. I tested Full House and other dice combinations to check if the scores were correct.

5. Challenges and Solutions

The final round was one of the most difficult parts. I needed to make sure every player got one final turn. I also had to keep track of all player scores and find the correct winner.

6. Retrospective

This homework helped me practice Java programming. I learned more about arrays, loops, methods, and game logic. I also got more experience with JUnit and testing my code.